Environmental Research 238 (2023) 117154

Contents lists available at ScienceDirect

Environmental Research

journal homepage: www.elsevier.com/locate/envres

Emergency department visits associated with wildfire smoke events in California, 2016–2019

Annie I. Chen [a], Keita Ebisu [a], Tarik Benmarhnia [b], Rupa Basu [a,*]

[a] Air and Climate Epidemiology Section, Office of Environmental Health Hazard Assessment, California Environmental Protection Agency, Oakland, CA, USA
[b] Scripps Institution of Oceanography, University of California San Diego, La Jolla, CA, USA

ARTICLE INFO

Handling Editor: Jose L Domingo

Keywords:
Wildfire smoke
Fine particulate matter
$PM_{2.5}$
Health impacts
Emergency department visits

ABSTRACT

Wildfire smoke has been associated with adverse respiratory outcomes, but the impacts of wildfire on other health outcomes and sensitive subpopulations are not fully understood. We examined associations between smoke events and emergency department visits (EDVs) for respiratory, cardiovascular, diabetes, and mental health outcomes in California during the wildfire season June–December 2016–2019. Daily, zip code tabulation area-level wildfire-specific fine particulate matter ($PM_{2.5}$) concentrations were aggregated to air basins. A "smoke event" was defined as an air basin-day with a wildfire-specific $PM_{2.5}$ concentration at or above the 98th percentile across all air basin-days (threshold = 13.5 μg/m^3). We conducted a two-stage time-series analysis using quasi-Poisson regression considering lag effects and random effects meta-analysis. We also conducted analyses stratified by race/ethnicity, age, and sex to assess potential effect modification. Smoke events were associated with an increased risk of EDVs for all respiratory diseases at lag 1 [14.4%, 95% confidence interval (CI): (6.8, 22.5)], asthma at lag 0 [57.1% (44.5, 70.8)], and chronic lower respiratory disease at lag 0 [12.7% (6.2, 19.6)]. We also found positive associations with EDVs for all cardiovascular diseases at lag 10. Mixed results were observed for mental health outcomes. Stratified results revealed potential disparities by race/ethnicity. Short-term exposure to smoke events was associated with increased respiratory and schizophrenia EDVs. Cardiovascular impacts may be delayed compared to respiratory outcomes.

1. Introduction

Wildfires have been increasing in frequency, duration, and severity in the past few decades largely due to climate change, particularly in the western United States (Westerling et al., 2006). In California, warmer temperatures, changes in precipitation regimes, drier conditions, and changes in seasonal prevailing winds have led to the rise of mega-fires and record-setting years of wildfires in recent years. Fifteen of the 20 most destructive wildfires recorded in California in the past century occurred after 2015, with the Camp Fire in 2018 being the most destructive and deadliest, resulting in over 18,000 structures destroyed and 85 deaths (CAL FIRE, 2022). Wildfires have devastating impacts on public health, ecosystems, buildings and infrastructure, and economies, costing the state billions of dollars each year (Feo, T.J. et al., 2020).

Wildfire smoke contains a mixture of particles and gases, with fine particulate matter ($PM_{2.5}$) as the primary pollutant of public health concern due to its toxicity and ability to penetrate deep into the lungs and reach other organs through the bloodstream (Liu et al., 2015; Wildfire Smoke: A Guide for Public Health Officials, 2021). Furthermore, $PM_{2.5}$ can disperse from the source and impact the air quality of communities not directly impacted by the fire. Ambient $PM_{2.5}$ (from multiple sources of emissions) has been shown to be causally linked to cardiovascular disease and strongly linked to other outcomes, including respiratory diseases and nervous system effects (US EPA, 2019). Consistent with the health effects of ambient $PM_{2.5}$, wildfire smoke has been associated with adverse respiratory outcomes in humans and in animals (Black et al., 2017; Carberry et al., 2022; Kim et al., 2018). However, epidemiologic associations between wildfire smoke and other health outcomes, such as cardiovascular diseases, mental health disorders, and adverse birth outcomes, have been mixed (Amjad et al., 2021; Chen et al., 2021; Moore et al., 2006).

There is growing evidence that wildfire-specific $PM_{2.5}$ may be more harmful than ambient $PM_{2.5}$ under non-wildfire conditions. Recent epidemiologic studies have found that wildfire-specific $PM_{2.5}$ may increase the risk of respiratory hospital visits by up to ten times more than that for non-wildfire ambient $PM_{2.5}$ (Aguilera et al., 2021a, 2021b).

* Corresponding author. 1515 Clay St. 16th Fl., Oakland, CA 94612, USA.
E-mail address: Rupa.Basu@oehha.ca.gov (R. Basu).

https://doi.org/10.1016/j.envres.2023.117154
Received 14 April 2023; Received in revised form 9 August 2023; Accepted 13 September 2023
Available online 15 September 2023
0013-9351/

Wildfire-specific $PM_{2.5}$ has also been shown to increase the risk of emergency department visits (EDVs) for headache disorders compared to the risk associated with overall $PM_{2.5}$ (Elser et al., 2023). These associations are supported by a toxicological study that observed an increased risk of inflammation in the lungs of mice after exposure to wildfire smoke compared to the risk after exposure to ambient $PM_{2.5}$ (Wegesser et al., 2009).

Furthermore, there is evidence that historically marginalized communities, including communities of color, are disproportionately exposed to wildfire smoke (Kondo et al., 2022; Méndez et al., 2020; Rappold et al., 2017). Yet, relatively few studies have investigated differential risk of health effects from wildfire smoke based on sociodemographic characteristics (Liu et al., 2017b). A meta-analysis of ten studies on wildfire smoke found a greater impact among females compared to males for asthma and chronic obstructive pulmonary disease and decreased risk of all respiratory outcomes for children compared to adults (Kondo et al., 2019). However, the investigators of the meta-analysis were unable to quantitatively assess effect modification by other sociodemographic characteristics, including race/ethnicity, due to insufficient numbers.

To address some of these gaps, we investigated associations between short-term exposure to smoke events, derived from wildfire-specific $PM_{2.5}$ concentrations, and EDVs for respiratory, cardiovascular, metabolic, and mental health outcomes in California from 2016 to 2019. We also conducted analyses stratified by race/ethnicity, sex, and age to assess whether some subpopulations are more susceptible to the health effects of smoke events compared to others. These results will contribute to our understanding of the health effects of wildfire smoke and help determine which subpopulations are more vulnerable. Identifying populations at increased risk of wildfire-related health impacts may guide public health officials and other government agencies in developing strategies for reducing wildfire smoke exposures and subsequent health impacts for those who are most affected.

2. Materials and methods

2.1. Exposure data

2.1.1. Wildfire-specific fine particulate matter ($PM_{2.5}$) concentrations

We obtained daily, zip code tabulation area (ZCTA)-level estimates of mean total and wildfire-specific $PM_{2.5}$ concentrations using an ensemble machine learning model (Aguilera et al., 2023). Briefly, three machine learning algorithms were combined to estimate daily, ZCTA-level total $PM_{2.5}$ concentrations, using about 50 predictor variables from satellite data, land use data, and meteorological data. Modeled total $PM_{2.5}$ concentrations showed high correlation with concentrations at all US Environmental Protection Agency's (USEPA) Air Quality System monitoring sites ($R^2 = 0.83$) and with previously published modeled concentrations (Aguilera et al., 2023; Di et al., 2019; Reid et al., 2021). Hazard Mapping System (HMS) smoke plume data were used to identify smoke-exposed zip code days. Counterfactual $PM_{2.5}$ concentrations (i.e. values expected in the absence of wildfire smoke) were estimated by first removing smoke-exposed zip code days from the dataset of total $PM_{2.5}$ concentrations and then imputing non-wildfire $PM_{2.5}$ concentrations using a fast implementation of chained random forest models. Wildfire-specific $PM_{2.5}$ concentrations were obtained by subtracting the counterfactual $PM_{2.5}$ concentrations from the total observed $PM_{2.5}$ concentrations. Although there is no gold standard for wildfire-specific $PM_{2.5}$ concentrations, this dataset was validated by plotting the distribution of wildfire-specific $PM_{2.5}$ concentrations by HMS smoke density: light, medium, and heavy.

California is divided into 15 air basins based on areas with similar geographic and meteorological features for air quality management purposes. Air basins were used as the unit of analysis in this study to obtain sufficient counts of EDVs to assess potential effect modification by race/ethnicity, age, and sex. Since air basins are defined based on similar meteorological and geographic characteristics, the characteristics of regional air pollutants are similar within air basins. Therefore, there is likely to be minimal exposure misclassification of regional air pollutants, such as wildfire-specific $PM_{2.5}$. Daily ZCTA-level concentrations were aggregated to the air basin-level using population-weighted averaging.

2.1.2. Definition of a smoke event

Concentrations of wildfire-specific $PM_{2.5}$ during months January through May were low (maximum concentration = 1.8 $\mu g/m^3$), so we restricted our analysis to months June through December. The distribution of wildfire-specific $PM_{2.5}$ during months June to December was still right-skewed, with a majority of air basin-days having zero exposure to wildfire-specific $PM_{2.5}$. Initial analyses using a continuous variable for wildfire-specific $PM_{2.5}$ revealed minimal associations with health outcomes. We therefore chose to focus on investigating exposure-outcome associations for days with high concentrations of wildfire-specific $PM_{2.5}$. To identify periods of high concentrations of wildfire-specific $PM_{2.5}$, we adapted a previously used exposure variable known as a "smoke wave" (Liu et al., 2017a). Since air basins were used as the unit of analysis, we defined a smoke event as an air basin-day with a wildfire-specific $PM_{2.5}$ concentration at or above the 98th percentile across all air basin-days during June–December 2016–2019 (threshold = 13.5 $\mu g/m^3$).

2.1.3. Wildfire perimeters

We obtained fire perimeters from the California Department of Forestry and Fire Protection's (CAL FIRE) Fire and Resource Assessment Program (https://frap.fire.ca.gov/mapping/gis-data/) (California Department of Forestry and Fire Protection, 2022). This data set consists of fire perimeters for wildfires and prescribed burns contributed by CAL FIRE, the United States Forest Service Region 5, the Bureau of Land Management, and the National Park Service. Fire perimeters were used to create a map showing the number of smoke events for each air basin overlaid with fires that occurred during the study period (Fig. 1).

2.1.4. Co-pollutant data

Monitored data for ozone (O_3), carbon monoxide (CO), and nitrogen dioxide (NO_2) were obtained from the USEPA Air Quality System Data Mart (US EPA, 2020). We included monitors with at least 90% daily coverage during the study period. For each ZCTA, we identified monitors that were within the same air basin and included monitors that were within a maximum distance of 5 km for CO and NO_2 and 10 km for O_3 to the population-weighted centroid. Daily, ZCTA-level concentrations were calculated using inverse distance weighting, based on the monitor distance to the population-weighted centroid. Daily, ZCTA-level concentrations were then aggregated to daily, air basin-level concentrations using population-weighted averaging. The exposure metrics for O_3, CO, and NO_2 were daily 8-h maximum (parts per billion, ppb), daily 1-h maximum (parts per million, ppm), and daily 1-h maximum (ppb), respectively.

2.1.5. Temperature data

Hourly temperature and relative humidity data were obtained from the USEPA, California Irrigation Management Information System (CIMIS), and the National Oceanic and Atmospheric Administration's National Centers for Environmental Information (NCEI) (National Oceanic and Atmospheric Administration, 2021; Office of Water Use Efficiency, CDWR, 2020; US EPA, 2020). We calculated mean daily apparent temperature, also known as the heat index, using formulas as previously described (Basu et al., 2008). Apparent temperature combines air temperature and relative humidity to represent what the temperature feels like.

We included monitors with at least 18 hourly measurements of temperature and relative humidity per day and with at least 90% daily coverage during the study period, December 1, 2015 through December

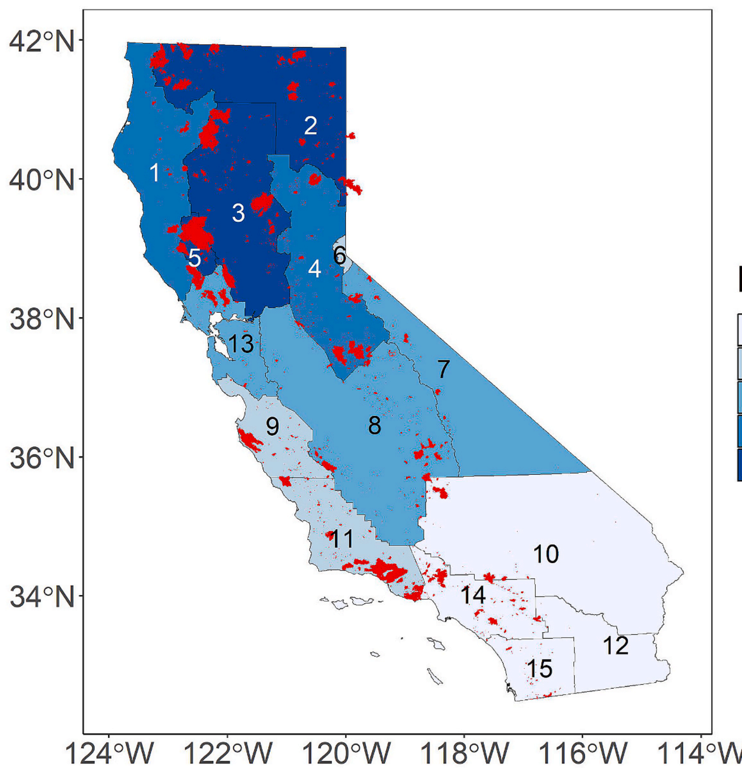

Fig. 1. Map of California indicating the number of smoke events by air basin, 2016-2019 June-December.
Fire perimeters are shown in red. Air basins: 1 – North Coast, 2 – Northeast Plateau, 3 – Sacramento Valley, 4 – Mountain Counties, 5 – Lake County, 6 – Lake Tahoe, 7 – Great Basin Valleys, 8 – San Joaquin Valley, 9 – North Central Coast, 10 – Mojave Desert, 11 – South Central Coast, 12 – Salton Sea, 13 – San Francisco Bay, 14 – South Coast, 15 – San Diego County. The four air basins without any smoke events (i.e. Mojave Desert, Salton Sea, South Coast, and San Diego County) were excluded from analyses.

31, 2019 (including one month before the start of the study period). For each ZCTA, we identified monitors that were within the same climate zone and air basin. The mean daily apparent temperature values for each ZCTA were calculated using inverse distance weighting, based on the monitor distance to the population-weighted centroid. Daily, ZCTA-level values were aggregated to daily, air basin-level values using population-weighted averaging.

2.2. Health outcome data

We prepared daily counts of unscheduled EDVs for years 2016–2019 using the Emergency Department Data and Patient Discharge Data datasets provided by the California Department of Health Care Access and Information (California Department of Health Care Access and Information (HCAI), 2021). The Patient Discharge Data were used to identify unscheduled acute or psychiatric inpatient hospitalizations that originated from the hospital's emergency department. We chose to include psychiatric care to fully capture EDVs for mental health disorders. For each encounter, we extracted the following variables: service or admission date, *International Classification of Disease, Tenth Revision, Clinical Modification* (ICD-10-CM) code for the principal diagnosis, race/ethnicity, age, sex, and residential zip code. Daily counts were summed by ZCTA and aggregated to the air basin level.

The principal diagnosis, reported using ICD-10-CM diagnosis codes, was used to group case counts into outcome categories for respiratory, cardiovascular, diabetes, and mental health outcomes. We used the Agency for Healthcare Research and Quality's (AHRQ) Clinical Classifications Software Refined (CCSR) tool, developed as part of the Healthcare Cost and Utilization Project, to group EDVs into clinically meaningful and mutually exclusive diagnosis categories ("Clinical Classifications Software Refined (CCSR) for ICD-10-CM Diagnoses," 2022). Version 2022-1 of the SAS mapping program provided by AHRQ (DXCCSR_Mapping_Program_v2022-1.sas) was used to map ICD-10-CM codes to their default CCSR category, based on principal diagnosis for inpatient data and the first-listed diagnosis for outpatient data. Version 2022-1 is compatible with ICD-10-CM codes from October 2015 through September 2022, which captures our entire study period. We reviewed the DXCCSR-Reference-File-v2022-1 to determine which default CCSR categories to group together for our final outcomes of interest and made manual adjustments for specific ICD-10-CM codes as needed (Table 1, Table S1). The final outcome categories included in this study were: all respiratory diseases, asthma, acute upper respiratory infections (AURI), chronic lower respiratory disease (CLRD), pneumonia, all cardiovascular diseases, acute myocardial infarction (AMI), cerebrovascular disease, dysrhythmia, ischemic heart disease (IHD), peripheral vascular disease (PVD), transient ischemic attack (TIA), diabetes, all mental health disorders, anxiety, mood disorders, schizophrenia, substance use disorders, and suicide/intentional self-harm (suicide) (Table 1).

Outcome counts were also stratified by race/ethnicity, age, and sex for subgroup analyses. The race/ethnicity categories were: Hispanic, non-Hispanic White ("White"), non-Hispanic Black ("Black"), and non-Hispanic Asian and Pacific Islander ("API"). Prior to 2019, the categories Asian and Native Hawaiian and Pacific Islander were combined into one category, so we combined those two categories for the data in 2019 to ensure consistency in the race/ethnicity coding. The age groups were as follows: 0–17 years old, 18–64 years old, and 65+ years old. For sex, we included males and females.

2.3. Population data

ZCTA-level population counts and sociodemographic data from the 2015–2019 5-year American Community Survey (ACS) were obtained using the tidycensus package in R and were used to calculate the total population and stratified population counts for each air basin (Walker and Herman, 2023). We used 2010 ZCTA population-weighted centroids and conducted a spatial join to assign each ZCTA to an air basin.

2.4. Statistical analysis

2.4.1. Main model

We conducted a two-stage time-series analysis to investigate associations between smoke events and EDVs for various causes. In the first

Table 1
Outcome categories and their corresponding Clinical Classifications Software Refined (CCSR) codes.

Outcome Group	CCSR codes
All respiratory diseases	RSP001-RSP017
Asthma	RSP009
Acute upper respiratory infections (AURI)	RSP001, RSP004, RSP006
Chronic lower respiratory disease (CLRD)	RSP008
Pneumonia	RSP002
All cardiovascular diseases	CIR001-CIR036, CIR038-CIR039
Acute myocardial infarction (AMI)	CIR009
Cerebrovascular disease	CIR020-CIR025
Dysrhythmia	CIR016-CIR018
Ischemic heart disease (IHD)	CIR009-CIR011
Peripheral vascular disease (PVD)	CIR026-CIR027, CIR029-CIR030, CIR032-CIR036, CIR039
Transient ischemic attack (TIA)	NVS012
Diabetes	END002-END003
All mental health disorders	MBD001-MBD014, MBD017-MBD027, MBD034
Anxiety	MBD005-MBD007, MBD011
Mood disorders	MBD002-MBD004, select ICD-10-CM codes from MBD026
Schizophrenia	MBD001
Substance use disorders	MBD017-MBD025, select ICD-10-CM codes in MBD026 and MBD034
Suicide	MBD012, MBD027, select ICD-10-CM codes in MBD034 and INJ075

stage, we ran time-series models for each air basin using a quasi-Poisson regression to account for overdispersion. In the second stage, we conducted random effects meta-analyses to combine air basin-specific estimates into an overall relative risk (RR) and corresponding 95% confidence intervals (CI), using the mixmeta package in R (Sera et al., 2019). The results are shown as percent change in risk, which was calculated as (RR – 1)*100%. To account for confounding, we adjusted for seasonal/long-term trends, holidays, day of the week, and mean apparent temperature at lag 0. We used lag 0 for mean apparent temperature because previous studies in California have reported positive associations between same-day mean apparent temperature and EDVs for various health outcomes (Basu et al., 2012, 2018). A natural cubic spline smoothing function of time was used to adjust for seasonal/long-term trends. The following holidays were combined into a single binary indicator variable: New Year's Eve, New Year's Day, Memorial Day, July 4th, Labor Day, Thanksgiving, and Christmas Day. We used six indicator variables for days of the week, with Sunday as the reference. We also included the log of the population as an offset to model rates rather than counts of EDVs, since population sizes varied across air basins and demographic groups. The main model was specified as follows:

$$Y_{ij} \sim \ Poisson\left(\mu_{ij}\right)$$

$$\begin{aligned}\log\left(\mu_{ij}\right) &= \alpha + \beta_1\left(smoke\ event\ indicator_{ij}\right) + \beta_2\left(mean\ apparent\ temp_{ij}\right) \\ &\quad + \beta_3\left(day\ of\ week_j\right) + \beta_4\left(holiday_j\right) + ns\left(date_j,\ df \right. \\ &\quad \left. = number\ of\ years*\frac{df}{yr}\right) + \log\ (population_i)\end{aligned}$$

where Y = EDV counts, i = air basin, j = observation date, df/yr = degrees of freedom per year.

Final model specifications were chosen by maximizing model fit. We chose the model with the smallest sum of the quasi-Akaike Information Criterion (qAIC) across the first-stage quasi-Poisson regression models. For the natural cubic spline, we tested between 1 and 6 degrees of freedom per year (df/yr). Four df/yr resulted in the best model fit for respiratory outcomes; for all other outcomes, we used two df/yr because of better model fit.

We also tested multiple lags and used the lag with the lowest total qAIC for each outcome. We looked at single-day lags and cumulative lags to investigate the effect of consecutive days of smoke events. For cumulative lags, each day in the lag period had to meet the threshold for a smoke event for the period to be considered a smoke event. We considered the following lags: same day (lag 0), previous day (lag 1), two days prior (lag 2), and cumulative lags of two days (lag 0_1) and three days (lag 0_2). Since previous studies have shown adverse respiratory outcomes within the first few days following wildfire smoke exposure, we did not test lags past 2 days for respiratory outcomes (Delfino et al., 2009; Reid et al., 2016b). For all other outcomes, we also considered longer lags: one week prior (lag 7), ten days prior (lag 10), and two weeks prior (lag 14). We used lag 0 for mean daily apparent temperature for all models.

We also assessed potential confounding by other air pollutants, including non-wildfire $PM_{2.5}$, O_3, CO, and NO_2. Some air basins were excluded from this analysis, due to insufficient monitor coverage across the state. For each air pollutant, we used the same subset of data and ran the main model as specified above with and without the co-pollutant in the model. Associations between smoke events and EDVs with and without the co-pollutant in the model were compared to determine whether there was potential confounding. We used the same lag for the co-pollutant and the main smoke event indicator for each outcome. Since these additional air pollutants are not thought to be associated with the concentration of wildfire-specific $PM_{2.5}$, we did not include these variables in the main model. In addition, we wanted to use the same set of covariates across all the health outcomes and opted to assess the robustness of the results by conducting sensitivity analyses with and without additional air pollutants in the model.

2.4.2. Effect modification

We conducted stratified analyses to assess potential effect modification by race/ethnicity, age group, and sex. We conducted a Z test of interaction to assess differences among subgroups, using Whites, 18–64 years old, and males as the reference groups, respectively (Altman, 2003). Some demographic and outcome combinations were excluded from stratified analyses due to insufficient numbers.

2.4.3. Sensitivity analyses

To assess the robustness of our results to different smoke event definitions, we used air basin-specific 98th percentile thresholds to identify smoke events for each air basin. We then compared associations based on the overall threshold, basin-specific thresholds for all 15 air basins, and basin-specific thresholds using the same subset of 11 basins as in the main analysis.

2.4.4. Health impact function

We applied a log-linear health impact function to estimate the number of EDVs attributable to smoke events. The effect estimate for the smoke event indicator was obtained from the second-stage meta-analysis that combined the air basin-specific estimates into an overall estimate for a given outcome (Table S2). For each outcome, attributable EDVs were calculated for each air basin separately, using air basin-specific values for baseline incidence rate, population, and number of smoke event days. Outcome-specific daily baseline incidence rates were calculated for each air basin by dividing the total number of EDVs for that outcome in a given air basin by the number of days in the study period (214 days/year * 4 years) and then dividing by the air basin population. Air basin-specific attributable EDVs were summed to obtain the total attributable EDVs on smoke event days for the 11 air basins included in this study. The following equation describes the health impact function used to estimate health burden on smoke event days: